

STRATEGIC SOURCING & PROCUREMENT

The Last Lecture

Lean Supply Chains in a Digital World



Quick question before we start:

What is the biggest waste you have personally seen in a supply chain?

KBS · KFUPM | MSc Supply Chain Management

Where We Have Been — and Where We End Today

✓ Strategic Sourcing

✓ Negotiation and Contract Management

→ Lean Supply Chain Management

✓ Supplier Selection & Evaluation

✓ Category Management

→ Digital Procurement Trends

✓ Supplier Relationships and Development

✓ Total Cost of Ownership

What We Cover in This Final Session

01

Inventory as Strategy

Asset or liability? Financial impact

02

Lean Supply Chain

JIT, pull systems, kanban

03

The 3-V Framework

Volume · Value · Velocity

04

Case Studies

Toyota, Amazon, Saudi Aramco

05

Simulation Exercise

You run the supply chain

06

Digital Frontier

AI · Automation · Platforms — the future of your career

Is Your Inventory an Asset — or a Liability?

Most firms treat inventory as a current asset on the balance sheet.
Lean thinking challenges this assumption entirely.

Traditional View

- Inventory = safety buffer
- Builds customer confidence
- Reflects strong purchasing

Lean View

- Inventory = frozen cash
- Hides real problems
- Reduces ROI and agility

The Hidden Price Tag of Holding Inventory

15.0%

Capital Tied Up

2.0%

Storage & Handling

1.2%

Obsolescence

1.05%

Taxes & Insurance

Total carrying cost averages 19–25% of inventory value per year

Calculation: Carrying Cost = Average Inventory × Unit Price × Annual Rate

Example: \$500,000 of inventory × 20% = \$100,000 per year just to hold it — before any stockouts or obsolescence.

How Inventory Directly Drives — or Kills — ROI

Same revenue (\$200M). Same product. One firm halves its inventory.

Watch what happens to return on investment:

Metric	Firm A (High Inventory)	Firm B (Lean)
Sales	\$200M	\$200M
Inventory Level	\$20M	\$10M
Inventory Turns	10×	20×
Profit Margin	6%	7%
Asset Turnover	1.56×	1.69×
ROI	9.4%	11.6%

 2.2 percentage points more ROI — from the same revenue. Inventory strategy is a financial strategy.

Right Reasons vs. Wrong Reasons for Inventory

✓ RIGHT Reasons

- Avoid supply disruptions (geopolitical, disasters)
- Support genuine demand variability
- Capture strategic quantity discounts
- Buffer for critical long-lead-time items

✗ WRONG Reasons (Fix These!)

- Poor supplier quality & delivery reliability
- Inaccurate demand forecasting systems
- Specifying custom parts for standard needs
- Inefficient, slow manufacturing processes

💡 Discussion: Which category do most Saudi manufacturers fall into — and why?

Lean: A Philosophy, Not a Toolbox

Lean is about shortening the time between customer order and delivery — by ruthlessly eliminating everything that **adds cost** but not value.



Flow

Inventory moves continuously —
no queuing, no waiting



Pull

Customer order triggers work —
not a forecast



Excellence

Zero **defects, zero waste** —
relentless improvement

JIT in Procurement: What It Actually Demands

JIT purchasing is not just 'order less' — it requires a fundamentally different supplier relationship model.



Zero-Defect Commitment

Incoming inspection eliminated — quality is built in at source



Collaborative Relationships

Single or dual sourcing with deep information sharing



EDI / Digital Integration

Real-time data exchange replaces paper POs and phone calls



Frequent Small Deliveries

Daily or even hourly deliveries at exact quantities needed



Stable Schedules

Suppliers need frozen windows to plan — no last-minute changes

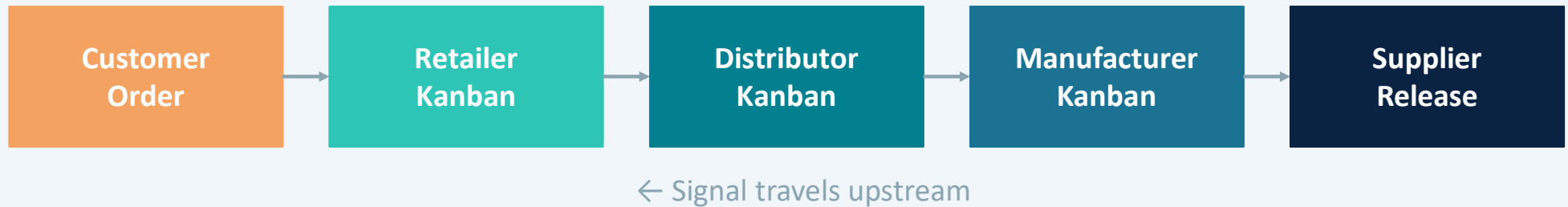


Geographic Proximity

Long supply chains undermine JIT — local sourcing preferred

Kanban: Letting Demand Do the Talking

Kanban is a visual signal system. No signal = no production = no waste. It turns the supply chain from forecast-pushed to demand-pulled.



Single-Card System

One card authorises one container of parts to move

Supplier Kanban

Card sent directly to the supplier to trigger replenishment

Electronic Kanban

ERP/EDI signal replaces physical card — used by Toyota, Bosch

Colour-Coded Bins

Red = order now, Yellow = monitor, Green = sufficient

Managing Inventory: The 3-V Model

Volume

? How MUCH inventory do we own?

 Key Measures

Total units, total weight, SKU count

 Actions

Forecasting accuracy, consignment inventory

Value

? What is it WORTH to us?

 Key Measures

Total \$, unit cost trends, sales-to-working-capital ratio

 Actions

Standardisation, part consolidation, leverage

Velocity

? How FAST does it move?

 Key Measures

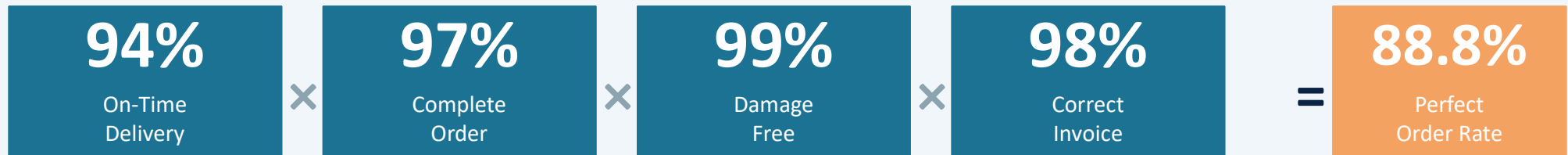
Inventory turns, throughput, order-to-cash cycle

 Actions

Lean practices, make-to-order production

The Perfect Order: The Ultimate Lean Goal

The perfect order means delivered on time · correct quantity · perfect condition · accurate documentation.
Every imperfect order costs you money — and costs you the customer.



 If each element is 94–99% perfect, the combined perfect order rate drops to under 89%. This is the multiplication problem of supply chain perfection.

MRP Systems

Time-phase material planning from production schedule

Inventory Planners

Dedicated roles managing inventory across the chain

DRP Systems

Coordinate finished goods across all distribution locations

Automated Tracking

RFID, barcode, IoT give real-time visibility

Real Companies. Real Results. Real Lessons.

We now look at three organisations that prove lean supply chain thinking is not theory — it is competitive advantage.



Toyota

Manufacturing | Japan / Global

The original lean supply chain —
and still the benchmark



Amazon

E-Commerce | USA / Global

How e-commerce redefined supply
chain velocity at scale



Saudi Aramco

Energy | Saudi Arabia

Procurement transformation in an
energy giant

Toyota: From 2 Weeks of Stock to 2 Hours

	GM Framingham (Traditional)	Toyota Takaoka (JIT Lean)
Assembly Hours / Vehicle	40.7 hrs	16.0 hrs
Defects per 100 Vehicles	130	45
Average Inventory	2 weeks	2 hours
Floor Space Used	Baseline	-40%



Key Lessons for Procurement:

Reduce supplier base → build trust → enable small frequent deliveries → eliminate incoming inspection → achieve JIT

Amazon: Velocity as Competitive Advantage

Amazon turned supply chain speed into a product. Prime delivery is not a logistics feature — it is the business model.

400M+

SKUs managed globally

< 2hrs

Same-day delivery in major cities

\$300B+

Annual procurement spend

75%

Orders fulfilled by robots in key centres

Inventory Positioning

Stock placed closest to where customers are — not where manufacturing is easy

Vendor-Managed Inventory

Suppliers get real-time sell-through data and are accountable for their own stock levels

Data-Driven Forecasting

ML models predict demand 18 months ahead, automatically triggering replenishment

Fulfillment Automation

Kiva robots (now Amazon Robotics) do warehouse picking — humans do exceptions only



Discussion: What can a Saudi retailer or manufacturer learn from Amazon's approach to inventory velocity?

Saudi Aramco: Lean Procurement at National Scale

With \$50B+ annual procurement spend, Aramco has transformed sourcing through its In-Kingdom Total Value Add (IKTVA) programme and digital procurement platforms.

IKTVA Programme

Mandates local content from suppliers — reducing global supply chain risk and building domestic capability. Target: 70% local content by 2030.

Ariba / SAP Integration



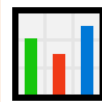
End-to-end digital procurement platform manages thousands of vendors, automates RFQs and POs, and tracks supplier performance in real time.

Vendor Managed Inventory



Key MRO suppliers hold consignment stock on-site at Aramco facilities — reducing Aramco's capital tied up while guaranteeing availability.

Supplier Scorecards



Quarterly KPI reviews cover delivery performance, quality, safety, and IKTVA contribution. Non-performers are put on improvement plans or delisted.



Vision 2030 connection: Lean procurement + local content = national supply chain resilience.

Your Turn: Run the Supply Chain

This is not a quiz. This is a decision-making exercise. You will face the same trade-offs real procurement managers face every day.

01

Your Role

You are the Procurement Manager of a Saudi FMCG company

02

The Scenario

Demand spike + key supplier delivers 3 days late

03

Your Decisions

How much buffer stock?
Single or dual source?
Lean or push?

04

The Reveal

We compare team decisions and see who survived — and who didn't

🕒 **Time: minutes | Teams | One page decision sheet**

The Perfect Storm Scenario

 You are: Procurement Director at AlMadar Foods, Riyadh — SAR 400M revenue, 60% of raw materials imported

The current date is the 3rd of Ramadan. Demand is 2.4× normal. Eid Al-Fitr is in 27 days.

Red Sea Disruption



Your primary sugar supplier in Egypt reports shipping delays of 8–12 days due to Houthi activity. Alternative routes add 15% cost.

Demand Spike



Modern trade orders up 180%. Hypermarkets threatening to delist you if out-of-stock rate exceeds 5% in Ramadan.

Cash Constraint



Finance has frozen working capital. You can order maximum SAR 8M additional inventory across all categories.

Spot Supplier Option



A broker offers sugar from India at 22% premium, FOB, 4-day lead time. Quality unverified. Take it or leave it?

Your Decision Framework — 15 Minutes

1. Buffer Stock Level

How many days of safety stock do you hold for sugar? 7 days / 14 days / 30 days? Justify using carrying cost vs stockout cost logic.

2. Spot Supplier

Accept the Indian broker offer at 22% premium? Yes / No / Partial quantity? What quality controls do you impose?

3. Allocation Priority

You cannot supply everyone. Do you prioritise: Hypermarkets (volume) / Pharmacies (margin) / Export orders (strategic)?

4. Communication Plan

Do you proactively inform your top 5 customers about potential delays? What is your risk message?

5. Lesson for Future

After Eid, what one structural change do you make to your sourcing strategy to prevent this situation next year?



Record your team's decisions. We debrief in 15 minutes.

What Actually Happened — and What It Teaches

Companies that survived the 2020–2024 disruptions (COVID, Red Sea, chip shortage) had these things in common:

01 • Dual Sourcing

Single-source dependency was the #1 reason for failure. Not cost — risk.

02 • Visibility Tools

Firms with real-time supply chain visibility recovered 2× faster.

03 • Pre-agreed Contracts

Spot market reliance cost 30–60% more. Framework contracts saved the day.

04 • Supplier Relationships

Suppliers prioritised customers who treated them as partners, not vendors.

05 • Cash Strategy

Firms with revolving credit for working capital avoided stockouts.

06 • Lean ≠ Fragile

Lean is resilient when paired with visibility and relationship depth.

Discussion: The Lean Paradox in Saudi Arabia

Saudi Arabia has world-class companies AND significant supply chain challenges simultaneously.

Q1 Is JIT realistic in Saudi Arabia given import dependency, geographic distances, and port congestion? What adaptations are needed?

Q2 Saudi Vision 2030 pushes for local content (IKTVA, NIDLP). Does this support or conflict with lean supply chain principles?

Q3 How should a Saudi company balance lean principles (low inventory) with geopolitical risk (Red Sea disruptions, sanctions)?

 Take 3 minutes to discuss with your neighbour. We'll hear two perspectives per question.

Discussion: You Are the CPO — What Do You Do?

You are the Chief Procurement Officer of a mid-size Saudi manufacturer. You have SAR 2M and 12 months to transform your supply chain.

Option A

Invest in supplier development — reduce your top-10 supplier defect rate from 4% to 0.5%. Lean from the source.

Option B

Deploy an e-procurement system (SAP Ariba / Jaggaer). Automate 70% of routine POs. Free up buyers for strategic work.

Option C

Implement vendor-managed inventory for your top-20 MRO items. Eliminate your stores department entirely.

Option D

Build a dual-source strategy for your top-5 critical materials. Slightly higher unit cost, massively lower risk.

 **Vote: Which option do you choose — and what is the one metric you track to know it worked?**

The Future of Procurement Is Already Here

This is the final chapter of your course — and the opening chapter of your career. The tools reshaping procurement will define who leads and who follows.



**AI & Machine
Learning**



**Autonomous
Procurement**



**Blockchain
Traceability**



**Predictive
Analytics**



Digital Twins



**Sustainability
Tech**

AI in Procurement: From Hype to Hard ROI

AI is not replacing procurement professionals — it is amplifying them.
The question is: will you be the one who uses it, or the one it replaces?



Demand Forecasting

ML models analyse POS data, social trends, economic indicators. Companies like Unilever predict demand 26 weeks ahead with 95%+ accuracy.



Supplier Risk Monitoring

Real-time monitoring of 10,000+ suppliers for financial distress, news signals, ESG violations, and geopolitical events.



Automated Sourcing

AI issues and evaluates RFQs for tail spend autonomously. Buyers review exceptions — routine is fully automated.



Spend Analytics

AI categorises millions of transactions in seconds. Identifies maverick spend, duplicate invoices, and savings opportunities humans miss.



Contract Intelligence

NLP reads and extracts key terms from thousands of contracts. Flags risks, renewal dates, and non-compliance clauses automatically.



Price Benchmarking

Continuously monitors market prices for your categories. Alerts when you are overpaying vs. market by >5%.

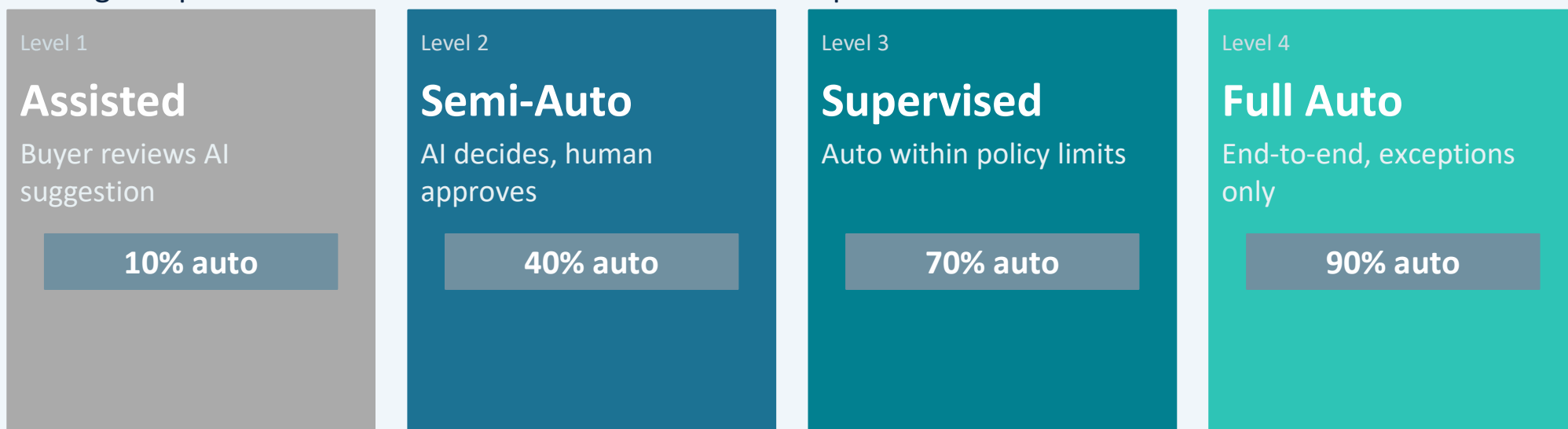


Platforms: Coupa, SAP Ariba, Jaggaer, Ivalua, GEP SMART — all now AI-native.

Autonomous & Touchless Procurement

Touchless procurement means a purchase goes from need → PO → receipt → invoice → payment with zero human intervention.

Leading companies now achieve 60–80% touchless on tail spend.

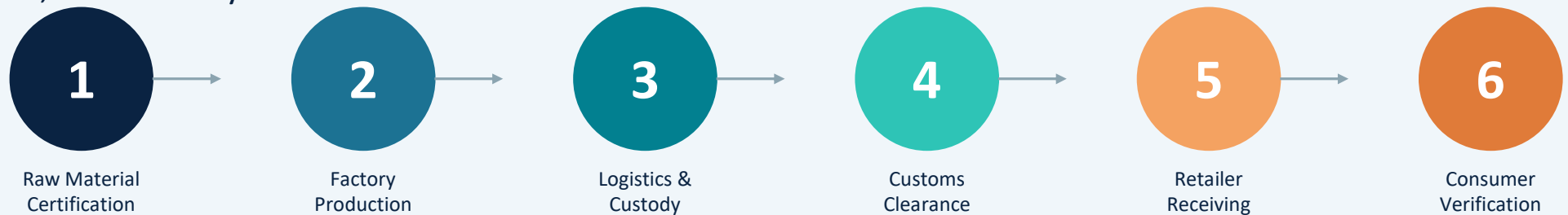


What gets automated first:

- MRO catalogue buying
- Invoice matching (3-way)
- Spot market commodity purchases
- Low-value services re-orders
- Travel & expense procurement

Blockchain: Transparency Across the Entire Supply Chain

Blockchain creates an immutable, shared ledger of every supply chain transaction — visible to all authorised parties, falsifiable by none.



Every step is timestamped and immutable on the shared ledger

Walmart (Food Safety)

Traces mango from farm to shelf in 2.2 seconds — was 6 days. Saves lives in contamination events.

De Beers (Diamonds)

Tracr platform tracks every diamond from mine to ring — no conflict diamonds in the chain.

Saudi Customs (ZATCA)

e-Invoicing mandates and blockchain-based customs clearance — procurement teams must be compliant.

Predictive Analytics & Digital Twins in SCM



Predictive Analytics

- Predicts supplier disruption 6–8 weeks before it happens using shipping data, financial signals, news feeds
- Identifies which SKUs will stock out in next 14 days — before your ERP system knows
- Optimises reorder points dynamically based on live demand vs. static safety stock formulas

Digital Twins

- A virtual replica of your entire supply chain — you can simulate disruptions before they happen
- Used by Unilever, P&G: test the impact of a port closure or a supplier failure in minutes
- NEOM Project uses digital twin of entire supply chain to plan mega-project logistics

Sustainability Is Now a Procurement Requirement

ESG compliance, Scope 3 emissions tracking, and supplier sustainability scoring are no longer optional. They are procurement KPIs.

Scope 1

Your own factory emissions

Direct control

Scope 2

Energy you purchase

Utility contracts

Scope 3

Your SUPPLY CHAIN emissions

Up to 80% of total footprint —
Procurement owns this

EcoVadis

Sustainability scorecards for 100,000+ suppliers.
Mandatory for EU supply chains.

Sourcemap

Maps your supply chain 3–4 tiers deep. Identifies child labour & environmental risk.

Watershed / Persefoni

Scope 3 emissions calculation from procurement spend data automatically.

Saudi SGX / Tadawul ESG

Saudi-listed companies must disclose supply chain ESG data from 2025.

The Modern Procurement Technology Stack

Know these platforms. They will be in every job description you read after graduation.

Source-to-Pay

SAP Ariba · **Coupa** · **Ivalua** · **Jaggaer**

End-to-end: sourcing → contracting → PO → invoice → payment

Spend Analytics

Sievo · **Spend HQ** · **Rosslyn Analytics**

Spend visibility, category benchmarking, savings tracking

Contract Management

Ironclad · **Icertis** · **ContractPodAi**

AI contract extraction, obligation tracking, renewal alerts

Supplier Management

GEP SMART · **Tealbook** · **Zycus**

Supplier onboarding, performance scoring, risk monitoring

Risk Intelligence

Resilinc · **Bindexis** · **riskmethods**

Real-time supply chain risk: supplier health, geopolitical, ESG

Saudi-Specific

Etimad · **Maroof** · **SFDA eProcurement**

Government procurement portals mandatory for Saudi public sector

The Procurement Professional of 2030

The procurement role is being redefined. Less transactional. More strategic, analytical, and technology-enabled.

Data Literacy

Read spend dashboards, run supplier analytics, interpret ML forecasts

Category Strategy

Go beyond price negotiation — own total cost, innovation, and risk for your category

Supplier Development

Develop capability in your supply base — teach lean, quality, and sustainability

Technology Fluency

Champion Ariba, Coupa, or equivalent. Be the one who trains others.

ESG & Compliance

Scope 3 reporting, supplier audits, ZATCA compliance — mandatory skills by 2027

Arabic + English + Data

Bilingual professionals who speak data are rare and extremely valuable in Saudi market

What This Course Has Built in You

Strategic Sourcing & Procurement is not about buying cheaper.
It is about creating competitive advantage through intelligent supply

1

Think Strategically

Every purchase decision has financial, operational, and relationship implications — own them all

2

Manage Total Cost

Price is what you pay. Cost is what you own. TCO, carrying cost, and quality cost are yours.

3

Build Relationships

Your supplier network is a strategic asset. Invest in it like you invest in technology.

4

Embrace Digital

The tools are here. The question is whether you will lead their adoption — or resist it.

5

Stay Curious

Procurement is one of the fastest-evolving functions in business. Never stop learning.

Thank You

"The supply chain is the business. Do it right, and everything else follows."

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Final Questions Worth Carrying Into Your Career:

- Is your organisation's inventory an asset — or a hidden liability?
- What one waste exists in your current or future workplace supply chain?
- Which digital tool will you master first — and how will you champion it?

Good luck. Make lean thinking a habit, not a course topic.